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1. Experiment Title:

Create a HTML form and use required JavaScript and events in the form.

2. Software/Tools Required:

Google Chrome (Web Browser), Visual Studio Code (Code Editor), Browser Developer Tools / Console, HTML, CSS, Javascript

3. Experiment Program Code:

Case Study 7.html

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Registration Form - Case Study</title>
    <link rel="stylesheet" href="Case Study 7.css">
    <link rel="preconnect" href="https://fonts.googleapis.com">
    <link rel="preconnect" href="https://fonts.gstatic.com" crossorigin>
    <link href="https://fonts.googleapis.com/css2?family=Inter:wght@300;400;500;600;700&display=swap"
rel="stylesheet">
</head>
<body>
    <div class="container">
        <h2>Case Study</h2>

        <form id="registrationForm" onsubmit="return validateForm(event)">
            <div class="mandatory-note">*Mandatory to fill</div>

            <div class="form-row">
                <label for="firstname">Firstname:</label>
                <div class="input-container">
                    <input type="text" id="firstname" name="firstname" placeholder="Enter first name">
                    <span class="required-star">*</span>
                </div>
            </div>
            <div class="error-msg" id="firstnameError"></div>

            <div class="form-row">
                <label for="lastname">Lastname:</label>
                <div class="input-container">
                    <input type="text" id="lastname" name="lastname" placeholder="Enter last name">
                </div>
            </div>
            <div class="error-msg" id="lastnameError"></div>

            <div class="form-row">
                <label>Birthday:</label>
                <div class="input-container dob-group">
                    <select id="dobDay" name="dobDay">
                        <option value="">Day</option>
                    </select>
                    <select id="dobMonth" name="dobMonth">
                        <option value="">Month</option>
                        <option value="1">Jan</option>
```

```
                        <option value="2">Feb</option>
                        <option value="3">Mar</option>
                        <option value="4">Apr</option>
                        <option value="5">May</option>
                        <option value="6">Jun</option>
                        <option value="7">Jul</option>
                        <option value="8">Aug</option>
                        <option value="9">Sep</option>
                        <option value="10">Oct</option>
                        <option value="11">Nov</option>
                        <option value="12">Dec</option>
                    </select>
                    <select id="dobYear" name="dobYear">
                        <option value="">Year</option>
                    </select>
                </div>
            </div>
            <div class="error-msg" id="dobError"></div>

            <div class="form-row">
                <label for="username">Username:</label>
                <div class="input-container">
                    <input type="text" id="username" name="username" placeholder="Enter username">
                    <span class="required-star">*</span>
                </div>
            </div>
            <div class="error-msg" id="usernameError"></div>

            <div class="form-row">
                <label for="email">E-mail:</label>
                <div class="input-container">
                    <input type="email" id="email" name="email" placeholder="Enter email address">
                </div>
            </div>
            <div class="error-msg" id="emailError"></div>

            <div class="form-row">
                <label for="website">Website:</label>
                <div class="input-container">
                    <input type="url" id="website" name="website" placeholder="https://example.com">
                </div>
            </div>
            <div class="error-msg" id="websiteError"></div>

            <div class="form-row">
                <label for="password">Password:</label>
                <div class="input-container">
                    <input type="password" id="password" name="password" placeholder="Enter password">
                    <span class="required-star">*</span>
                </div>
            </div>
            <div class="error-msg" id="passwordError"></div>

            <div class="form-row">
                <label for="rePassword">Re-password:</label>
                <div class="input-container">
                    <input type="password" id="rePassword" name="rePassword" placeholder="Confirm
password">
                    <span class="required-star">*</span>
                </div>
            </div>
            <div class="error-msg" id="rePasswordError"></div>

            <div class="checkbox-row">
                <input type="checkbox" id="terms" name="terms">
                <label for="terms">I agree to the terms & conditions.</label>
            </div>
            <div class="error-msg" id="termsError"></div>

            <div class="button-row">
```

```
                <button type="submit" id="signUpBtn">Sign up</button>
            </div>

            <div id="successMsg" class="success-msg"></div>
        </form>

        <footer style="text-align: center; margin-top: 30px; font-size: 14px; color: gray; border-top: 1px
solid lightgray; padding-top: 15px;">
            Made by Rohan Laharwani PRN - 24070521055
        </footer>
    </div>

    <script src="Case Study 7.js"></script>
</body>
</html>
```

Case Study 7.css

```
* {
    box-sizing: border-box;
}

body {
    background-color: whitesmoke;
    color: black;
    font-family: 'Inter', sans-serif;
    margin: 0;
    padding: 40px 20px;
    display: flex;
    justify-content: center;
}

.container {
    width: 100%;
    max-width: 560px;
    background-color: white;
    padding: 35px 40px;
    border: 1px solid lightgray;
    border-radius: 12px;
    box-shadow: 0 4px 15px lightgray;
}

h2 {
    color: darkblue;
    text-align: center;
    margin-top: 0;
    margin-bottom: 24px;
    font-size: 24px;
}

.mandatory-note {
    color: red;
    font-weight: 600;
    font-size: 13px;
    margin-bottom: 18px;
}

.form-row {
    display: flex;
    align-items: center;
    margin-bottom: 12px;
}

.form-row label {
    width: 120px;
    font-size: 14px;
    font-weight: 500;
    color: black;
```

```
}

.input-container {
    flex: 1;
    display: flex;
    align-items: center;
    position: relative;
}

.input-container input,
.input-container select {
    width: 100%;
    padding: 8px 12px;
    border: 1px solid lightgray;
    border-radius: 6px;
    font-size: 14px;
    font-family: inherit;
    outline: none;
    background-color: white;
}

.input-container input:focus,
.input-container select:focus {
    border-color: royalblue;
}

.required-star {
    color: red;
    margin-left: 8px;
    font-weight: bold;
    font-size: 15px;
    user-select: none;
}

.dob-group {
    display: flex;
    gap: 8px;
}

.dob-group select {
    flex: 1;
    cursor: pointer;
}

.checkbox-row  {
    display: flex;
    align-items: center;
    margin-top: 14px;
    margin-bottom: 4px;
    gap: 8px;
}

.checkbox-row input[type="checkbox"] {
    width: 16px;
    height: 16px;
    cursor: pointer;
}

.checkbox-row label {
    font-size: 13px;
    color: black;
    cursor: pointer;
}

.error-msg {
    color: red;
    font-size: 12px;
    margin-left: 120px;
    margin-bottom: 8px;
```

```
    min-height: 14px;
}

.button-row {
    margin-top: 18px;
}

#signUpBtn {
    padding: 10px 24px;
    background-color: royalblue;
    color: white;
    border: none;
    border-radius: 6px;
    font-size: 14px;
    font-weight: 600;
    cursor: pointer;
}

#signUpBtn:hover {
    background-color: darkblue;
}

.success-msg {
    display: none;
    margin-top: 20px;
    padding: 12px 16px;
    background-color: lightgreen;
    color: darkgreen;
    border: 1px solid green;
    border-radius: 6px;
    font-size: 14px;
    text-align: center;
    font-weight: 500;
}
```

Case Study 7.js

```
let dobDay = document.getElementById("dobDay");
for (let i = 1; i <= 31; i++) {
    let opt = document.createElement("option");
    opt.value = i;
    opt.text = i;
    dobDay.appendChild(opt);
}

let dobYear = document.getElementById("dobYear");
let currentYear = new Date().getFullYear();
for (let y = currentYear; y >= 1950; y--) {
    let opt = document.createElement("option");
    opt.value = y;
    opt.text = y;
    dobYear.appendChild(opt);
}

function clearErrors() {
    let errors = document.getElementsByClassName("error-msg");
    for (let i = 0; i < errors.length; i++) {
        errors[i].innerHTML = "";
    }
}

let allInputs = document.querySelectorAll("input[type='text'], input[type='email'], input[type='url'],
input[type='password']");

for (let i = 0; i < allInputs.length; i++) {
    allInputs[i].addEventListener("focus", function() {
        this.style.borderColor = "royalblue";
    });
```

```
    allInputs[i].addEventListener("blur", function() {
        this.style.borderColor = "lightgray";
    });
}

document.getElementById("terms").addEventListener("change", function() {
    if (this.checked) {
        document.getElementById("termsError").innerHTML = "";
    }
});

function validateForm(event) {
    if (event) {
        event.preventDefault();
    }

    clearErrors();
    let isValid = true;

    let firstname = document.getElementById("firstname").value.trim();
    if (firstname === "") {
        document.getElementById("firstnameError").innerHTML = "Firstname is required.";
        isValid = false;
    } else if (!/^[a-zA-Z]+$/.test(firstname)) {
        document.getElementById("firstnameError").innerHTML = "Only alphabets are allowed.";
        isValid = false;
    }

    let lastname = document.getElementById("lastname").value.trim();
    if (lastname !== "" && !/^[a-zA-Z]+$/.test(lastname)) {
        document.getElementById("lastnameError").innerHTML = "Only alphabets are allowed.";
        isValid = false;
    }

    let username = document.getElementById("username").value.trim();
    if (username === "") {
        document.getElementById("usernameError").innerHTML = "Username is required.";
        isValid = false;
    } else if (username.length < 4) {
        document.getElementById("usernameError").innerHTML = "Username must be at least 4 characters.";
        isValid = false;
    }

    let email = document.getElementById("email").value.trim();
    let emailPattern = /^[^\s@]+@[^\s@]+\.[^\s@]+$/;
    if (email !== "" && !emailPattern.test(email)) {
        document.getElementById("emailError").innerHTML = "Please enter a valid email address.";
        isValid = false;
    }

    let website = document.getElementById("website").value.trim();
    let urlPattern = /^(https?:\/\/)?([\da-z.-]+)\.([a-z.]{2,6})([\/\w .-]*)*\/?$/;
    if (website !== "" && !urlPattern.test(website)) {
        document.getElementById("websiteError").innerHTML = "Please enter a valid website URL.";
        isValid = false;
    }

    let password = document.getElementById("password").value;
    if (password === "") {
        document.getElementById("passwordError").innerHTML = "Password is required.";
        isValid = false;
    } else if (password.length < 6) {
        document.getElementById("passwordError").innerHTML = "Password must be at least 6 characters.";
        isValid = false;
    }

    let rePassword = document.getElementById("rePassword").value;
    if (rePassword === "") {
```

```
        document.getElementById("rePasswordError").innerHTML = "Please confirm your password.";
        isValid = false;
    } else if (password !== rePassword) {
        document.getElementById("rePasswordError").innerHTML = "Passwords do not match.";
        isValid = false;
    }

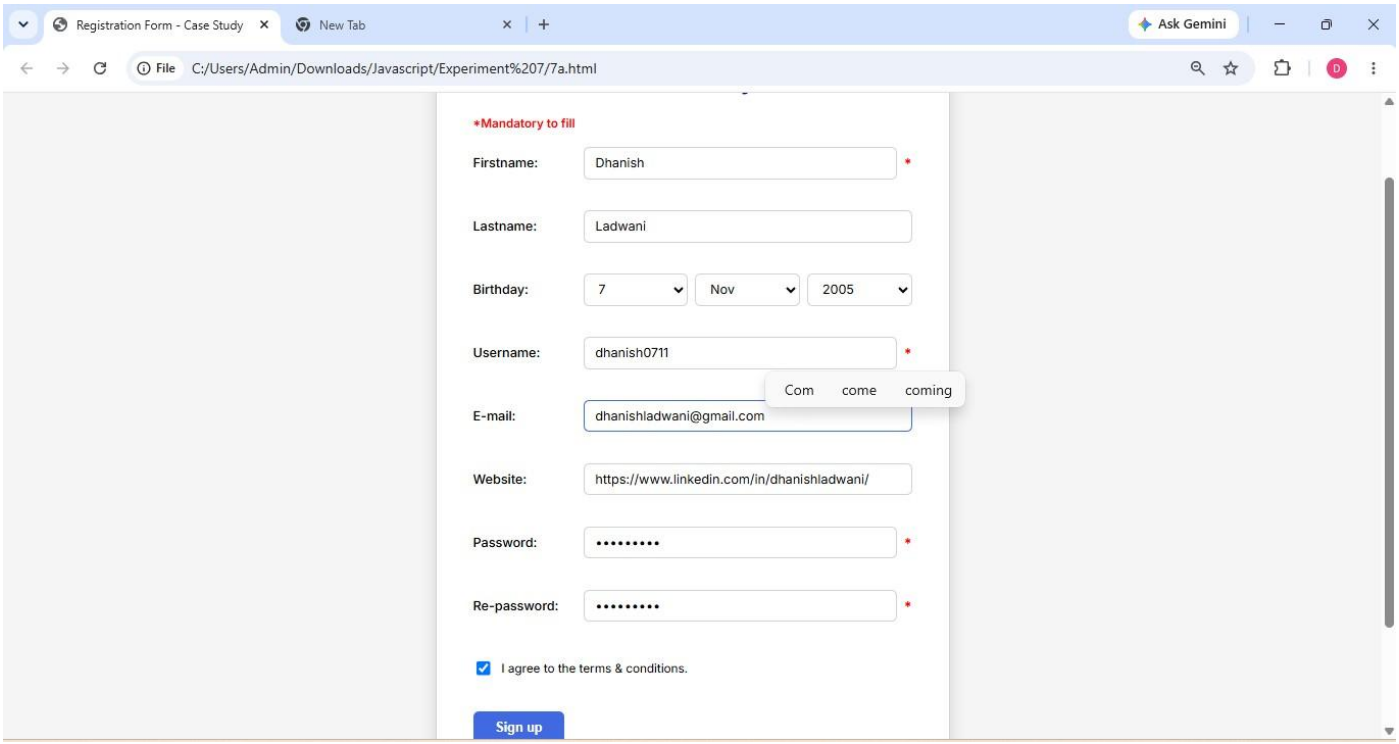
    let terms = document.getElementById("terms").checked;
    if (!terms) {
        document.getElementById("termsError").innerHTML = "You must agree to the terms & conditions.";
        isValid = false;
    }

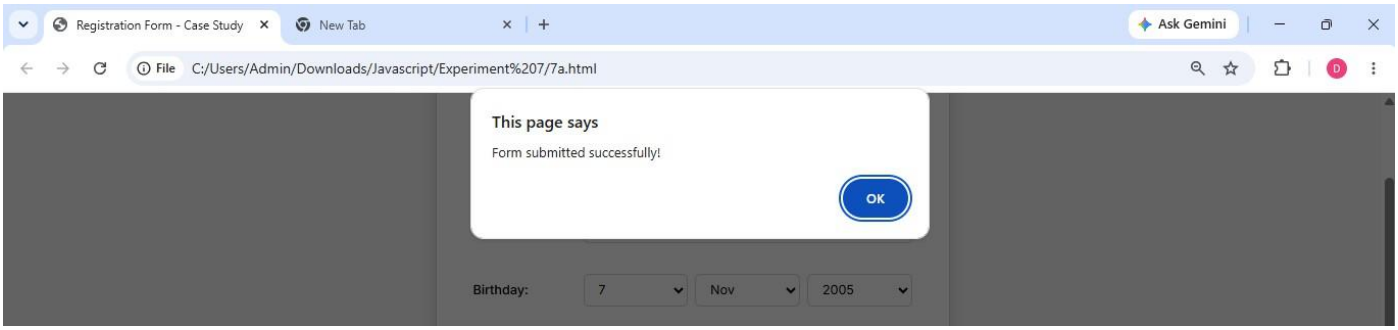
    if (isValid) {
        let successMsg = document.getElementById("successMsg");
        successMsg.style.display = "block";
        successMsg.innerHTML = "Registration Successful for <b>" + username + "</b>!";
        alert("Form submitted successfully!");
    }

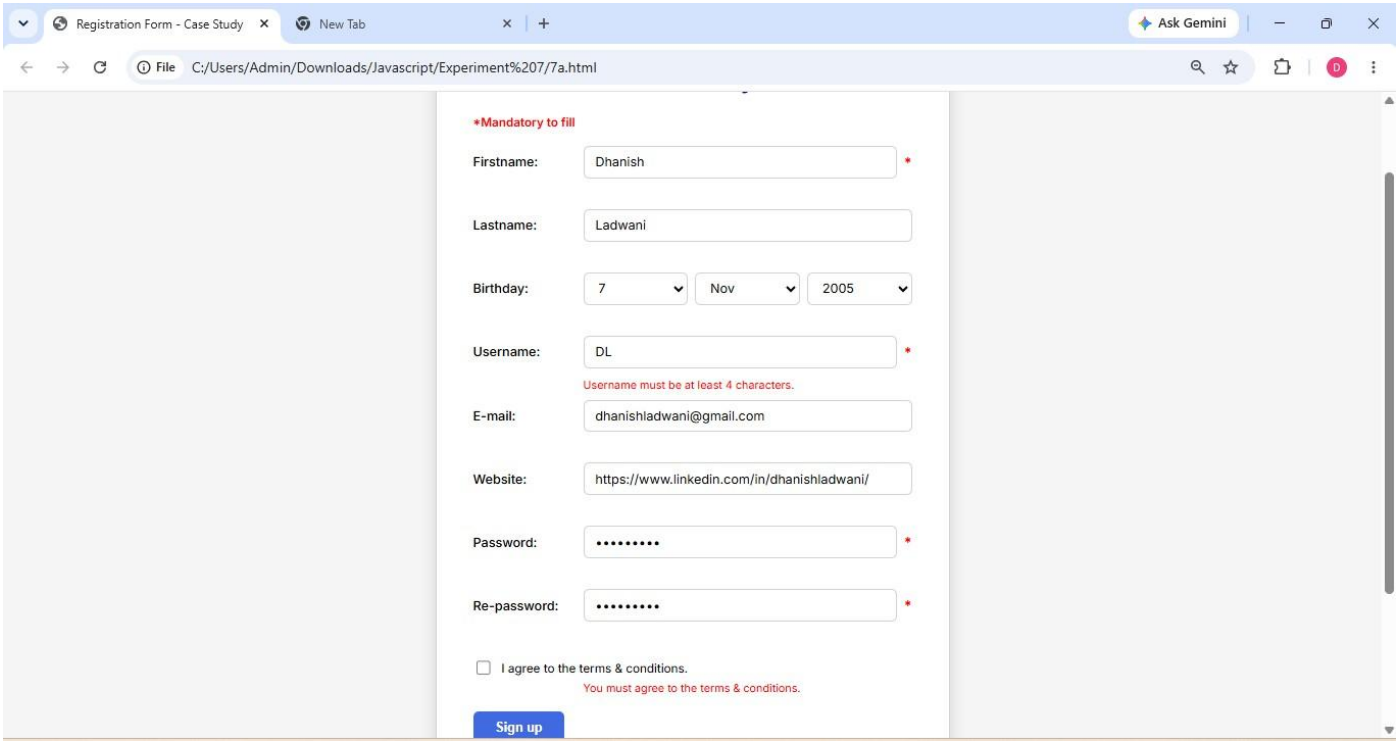
    return false;
}

document.getElementById("registrationForm").addEventListener("submit",  validateForm);
```

4. Output:


Registration Form - Case Study
New Tab
Ask Gemini
File C:/Users/Admin/Downloads/Javascript/Experiment%207/7a.html
*Mandatory to fill
Firstname: Dhanish *
Lastname: Ladwani
Birthday: 7 Nov 2005
Username: dhanish0711 *
Com come coming
E-mail: dhanishladwani@gmail.com
Website: https://www.linkedin.com/in/dhanishladwani/
Password: ••••••••• *
Re-password: ••••••••• *
I agree to the terms & conditions.
Sign up


Registration Form - Case Study
New Tab
Ask Gemini
File C:/Users/Admin/Downloads/Javascript/Experiment%207/7a.html
This page says
Form submitted successfully!
OK
Birthday: 7 Nov 2005


Registration Form - Case Study
New Tab
Ask Gemini
File C:/Users/Admin/Downloads/Javascript/Experiment%207/7a.html
*Mandatory to fill
Firstname: Dhanish *
Lastname: Ladwani
Birthday: 7 Nov 2005
Username: DL *
Username must be at least 4 characters.
E-mail: dhanishladwani@gmail.com
Website: https://www.linkedin.com/in/dhanishladwani/
Password: ••••••••• *
Re-password: ••••••••• *
I agree to the terms & conditions.
You must agree to the terms & conditions.
Sign up

5. Result/Conclusion:

A comprehensive registration form was developed with real-time field validation, dynamically generated date dropdowns, and interactive error styling that highlights invalid fields as the user types. This case study reinforced advanced form validation, DOM error handling and dynamic UI feedback using JavaScript.